

MIT OpenCourseWare

14.16

Strategy and Information 14.16 — Strategy and Information May 2026

EXTREMELY HARD – MIT LEVEL**Time Limit: 3 hours****Total Marks: 100**

1. (4 points) Analyze the limitations of standard approaches to Adverse Selection when applied to edge cases in 14.16. Construct explicit counterexamples showing where intuitive reasoning fails.
2. (2 points) Analyze the limitations of standard approaches to Moral Hazard when applied to edge cases in 14.16. Construct explicit counterexamples showing where intuitive reasoning fails.
3. (5 points) Construct an original proof-level problem involving Screening for 14.16 (Strategy and Information) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
4. (3 points) Construct an original proof-level problem involving Signaling for 14.16 (Strategy and Information) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
5. (3 points) Construct an original proof-level problem involving Reputation for 14.16 (Strategy and Information) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
6. (4 points) Analyze the limitations of standard approaches to Contracts when applied to edge cases in 14.16. Construct explicit counterexamples showing where intuitive reasoning fails.
7. (3 points) Construct an original proof-level problem involving Adverse Selection for 14.16 (Strategy and Information) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
8. (7 points) Construct an original proof-level problem involving Moral Hazard for 14.16 (Strategy and Information) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

9. (5 points) Construct an original proof-level problem involving Screening for 14.16 (Strategy and Information) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
10. (5 points) Analyze the limitations of standard approaches to Signaling when applied to edge cases in 14.16. Construct explicit counterexamples showing where intuitive reasoning fails.
11. (5 points) Construct an original proof-level problem involving Reputation for 14.16 (Strategy and Information) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
12. (7 points) Construct an original proof-level problem involving Contracts for 14.16 (Strategy and Information) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
13. (2 points) Analyze the limitations of standard approaches to Adverse Selection when applied to edge cases in 14.16. Construct explicit counterexamples showing where intuitive reasoning fails.
14. (3 points) Analyze the limitations of standard approaches to Moral Hazard when applied to edge cases in 14.16. Construct explicit counterexamples showing where intuitive reasoning fails.
15. (4 points) Construct an original proof-level problem involving Screening for 14.16 (Strategy and Information) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
16. (3 points) Prove the fundamental theorem concerning Signaling in the context of 14.16 (Strategy and Information). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
17. (7 points) Prove the fundamental theorem concerning Reputation in the context of 14.16 (Strategy and Information). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
18. (4 points) Prove the fundamental theorem concerning Contracts in the context of 14.16 (Strategy and Information). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
19. (4 points) Prove the fundamental theorem concerning Adverse Selection in the context of 14.16 (Strategy and Information). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

20. (3 points) Prove the fundamental theorem concerning Moral Hazard in the context of 14.16 (Strategy and Information). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
21. (4 points) Construct an original proof-level problem involving Screening for 14.16 (Strategy and Information) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
22. (5 points) Construct an original proof-level problem involving Signaling for 14.16 (Strategy and Information) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
23. (3 points) Prove the fundamental theorem concerning Reputation in the context of 14.16 (Strategy and Information). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
24. (3 points) Construct an original proof-level problem involving Contracts for 14.16 (Strategy and Information) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
25. (2 points) Construct an original proof-level problem involving Adverse Selection for 14.16 (Strategy and Information) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

Solutions

Solution 1:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Adverse Selection. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 2:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Moral Hazard. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 3:

Solution approach: First recall the definitions and key theorems related to Screening from 14.16. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 4:

Solution approach: First recall the definitions and key theorems related to Signaling from 14.16. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 5:

Solution approach: First recall the definitions and key theorems related to Reputation from 14.16. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 6:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Contracts. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 7:

Solution approach: First recall the definitions and key theorems related to Adverse Selection from 14.16. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 8:

Solution approach: First recall the definitions and key theorems related to Moral Hazard from 14.16. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 9:

Solution approach: First recall the definitions and key theorems related to Screening from 14.16. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 10:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Signaling. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 11:

Solution approach: First recall the definitions and key theorems related to Reputation from 14.16. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 12:

Solution approach: First recall the definitions and key theorems related to Contracts from 14.16. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 13:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Adverse Selection. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 14:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Moral Hazard. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 15:

Solution approach: First recall the definitions and key theorems related to Screening from 14.16. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 16:

The proof follows from the definitions and properties established in 14.16. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

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The proof follows from the definitions and properties established in 14.16. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 21:

Solution approach: First recall the definitions and key theorems related to Screening from 14.16. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 22:

Solution approach: First recall the definitions and key theorems related to Signaling from 14.16. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 23:

The proof follows from the definitions and properties established in 14.16. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 24:

Solution approach: First recall the definitions and key theorems related to Contracts from 14.16. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 25:

Solution approach: First recall the definitions and key theorems related to Adverse Selection from 14.16. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.